Predicting Fetal Distress

Classifying Fetal Health from Cardiotocography Data

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We are data scientists with backgrounds in product management, fintech, and data analytics.

The Problem

During pregnancy, Cardiotocography (CTG) monitors the baby's heart rate and the mother's contractions to check whether the baby is in distress.

The challenge: interpreting CTG traces is subjective. Different clinicians can read the same trace and reach different conclusions. This inconsistency can lead to two problems:

- Over-intervention -- unnecessary procedures on healthy pregnancies
- **Under-intervention -- missing a baby in distress who needs urgent help**

Our question:

Can we build a system that automatically classifies fetal health as Normal, Suspected, or Abnormal from CTG measurements -- providing consistent, data-driven assessments to support clinical decisions?

The Data

What the data contains

2,126 CTG records, each capturing:

Fetal heart rate patterns

Baseline heart rate, accelerations, decelerations

Variability indicators

Short-term and long-term fluctuations in heart rate

Uterine contractions and fetal movement

Heart rate distribution statistics

Width, peaks, mode, variance of the heart rate histogram

The class imbalance challenge

Normal: 77.8% (1,655 records)

The vast majority of pregnancies are healthy.

Suspected: 13.9% (295 records)

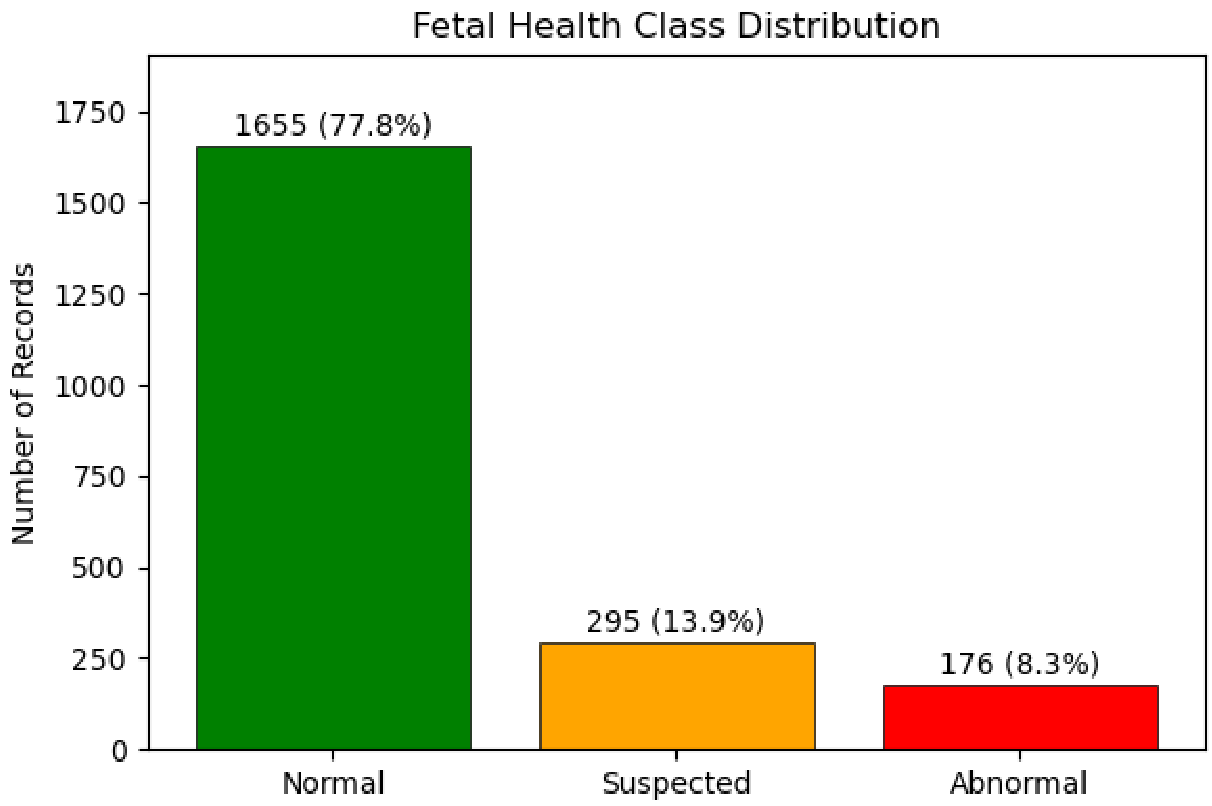
Cases warranting closer monitoring.

Abnormal: 8.3% (176 records)

Genuine fetal distress -- rare but critical to detect.

A model that always predicts 'Normal' would be 78% accurate but would miss every case of fetal distress.

Fetal Health Class Distribution



Our dataset has a class imbalance

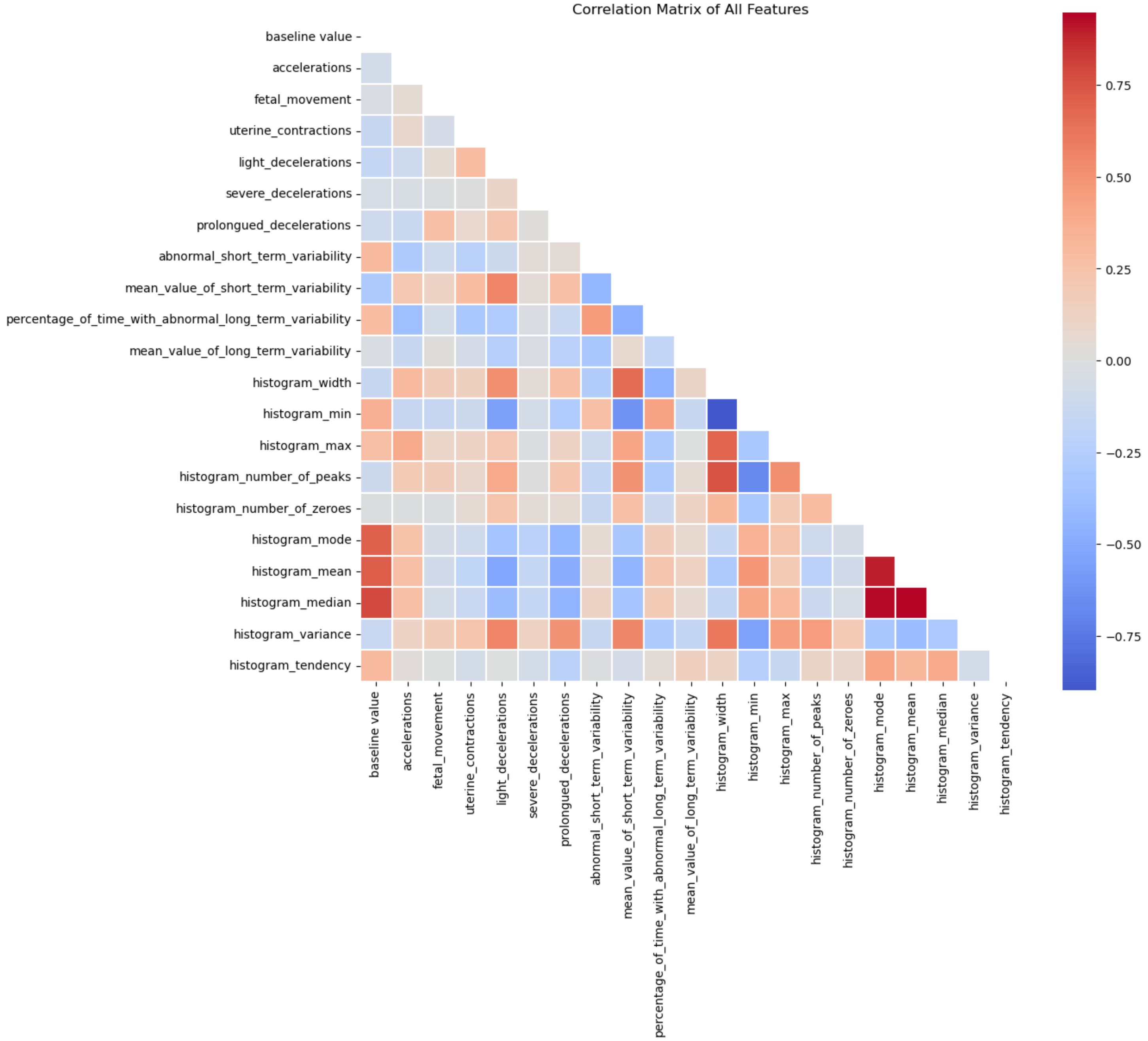
This has important implications for modelling:

- A naive model that predicts "Normal" for every case would achieve roughly 78% accuracy -- our models must substantially beat this baseline.

- Accuracy alone is misleading. We will use macro F1 as the primary comparison metric, which weights all three classes equally.

- We pay special attention to recall on the Abnormal class (Class 3), because in a clinical setting, missing a case of fetal distress is far more dangerous than a false alarm.

Correlation matrix of all features



- - Histogram features (mode, mean, median) are near-duplicates of each other -- we dropped the redundant ones
- - Clinical features (accelerations, decelerations, variability) are not correlated -- they each capture something different about fetal health
- - This analysis guided which features we kept and which we removed before modelling

Why Not All Errors Are Equal

In most prediction problems, errors in either direction are roughly equal. In healthcare, they are not.

False alarm (false positive)

A healthy baby is flagged for further testing. The mother undergoes an extra scan or monitoring session. Outcome: minor inconvenience and some anxiety, but no harm to mother or baby.

Cost: Low

Missed distress (false negative)

A baby in distress is classified as normal. No intervention occurs. The distress goes undetected, potentially leading to serious complications during labour and delivery.

Cost: Potentially life-threatening

This asymmetry shapes our entire approach: we optimise for catching Abnormal cases, even if it means a few more false alarms.

Our Approach

1. Logistic Regression

   A simple starting point to establish a baseline and confirm the data contains learnable patterns.

2. Support Vector Machine

   Tests whether non-linear boundaries between classes improve on the baseline.

3. Random Forest

   Combines many decision trees to detect complex patterns in the CTG measurements

4. Gradient Boosting

   Builds trees that learn from each other's mistakes -- typically the strongest approach for structured data.

5. Gradient Boosting + SMOTE

   Same model, but we first balanced the training data by creating synthetic examples of Suspected and Abnormal cases -- the classes every model struggled with.

6. Stacked Ensemble + SMOTE

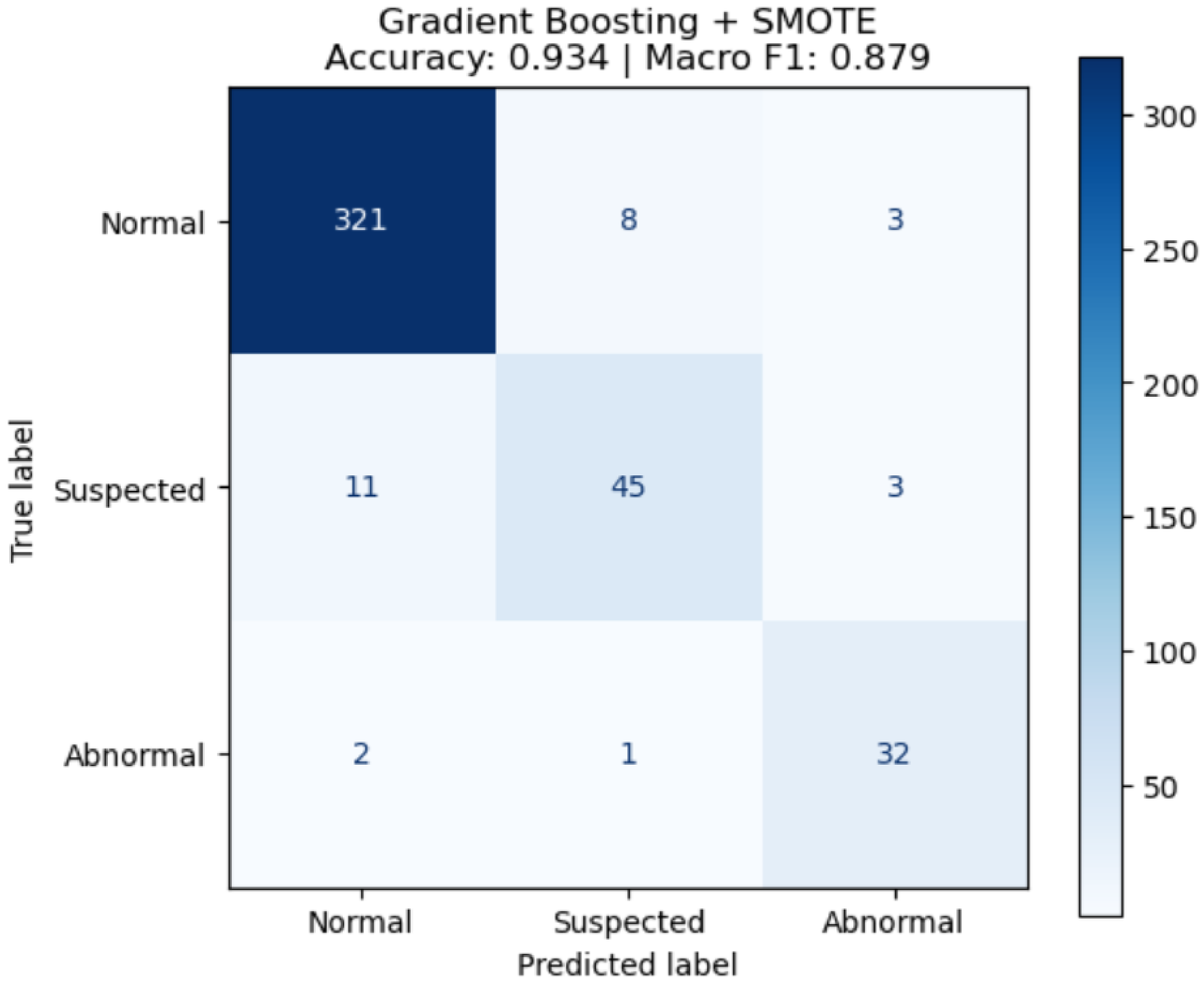
   We combined Random Forest and Gradient Boosting, using a simple model to learn the best way to merge their predictions. This tests whether combining models outperforms a single well-tuned model.

Results

Model	Accuracy	Balanced Score (F1)	Abnormal Detection
Logistic Regression	87.6%	0.764	23 of 35 (65.7%)
Support Vector Machine	89.0%	0.769	23 of 35 (65.7%)
Random Forest	92.5%	0.847	28 of 35 (80.0%)
Gradient Boosting	93.2%	0.860	28 of 35 (80.0%)
GB + SMOTE (final model)	**93.4%**	**0.879**	**32 of 35 (91.4%)**
Stacked Ensemble + SMOTE	92.5%	0.856	30 of 35 (85.7%)
Baseline (always predict Normal)	77.8%	--	0 of 35 (0%)

Our final model detects 32 out of 35 cases of fetal distress, missing only 2.

GB + SMOTE Confusion Matrix



What Matters Most: Which Measurements Predict Distress?

Not all CTG measurements are equally informative. Our models consistently identified the same top predictors:

Strongest predictors

Heart rate variability

Abnormal short-term and long-term variability were the top-ranked features. Reduced variability is a well-established clinical sign of fetal distress.

Prolonged decelerations

Sustained drops in heart rate that can indicate cord compression or placental problems. Consistently in the top five.

Accelerations

Their presence is a reassuring sign. Their absence helps identify at-risk cases.

Weakest predictors

Severe decelerations

Clinically important but extremely rare in the data -- most records have zero. The model did not have enough examples to learn from.

Light decelerations

Generally considered benign in clinical practice, so their low ranking is consistent with medical knowledge.

Fetal movement

Can be unreliable in CTG monitoring and contributed little to predictions.

The takeaway for clinicians: when reviewing a CTG trace, focus on heart rate variability and prolonged decelerations -- these are the most informative indicators of fetal well-being.

Key Takeaways

Automated CTG classification works.

Our final model achieves 93.4% accuracy and catches 91.4% of Abnormal cases. It substantially outperforms a baseline that would miss all cases of fetal distress.

The most dangerous errors have been minimised.

Only 2 out of 35 Abnormal cases were misclassified as Normal -- the most clinically dangerous error. The model errs on the side of caution, which is the right trade-off in healthcare.

Heart rate variability is the strongest signal.

Across all models, variability indicators dominated feature importance. This aligns with established clinical knowledge and reinforces where clinicians should focus when interpreting CTG traces.

Addressing class imbalance was essential.

Abnormal cases make up only 8.3% of the data. Without SMOTE, models detected 80% of Abnormal cases. With SMOTE, detection rose to 91.4% -- a meaningful improvement that could prevent harm in real clinical settings.

Limitations and Future Work

Current limitations

A decision-support tool, not a replacement for clinical judgment. Final decisions must remain with healthcare professionals.

Relatively small dataset (2,126 records). Performance should be validated on larger, external datasets from different hospitals and populations.

Static measurements only. Real CTG monitoring captures changes over time, which this dataset does not reflect.

Severe decelerations -- clinically important -- are too rare in the data for the model to learn from effectively.

Where to go next

Validate on external data from different hospitals and regions before any clinical deployment.

Add time-series CTG data to capture how heart rate patterns change during monitoring -- developing distress may look different from sustained distress.

Include patient context -- gestational age, maternal health, labour stage -- to test whether clinical context improves predictions.

Thank You

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Questions?